

Thaanvi Sudarsan Meda

Charlotte, NC — +1 (704) 241-6430 — medathanvi@gmail.com

LinkedIn — GitHub

Mobile Engineer with **3+ years of combined industry and research experience** building scalable mobile applications and AI-powered software solutions. Skilled in **Swift, UIKit, REST API integration, Agile development, testing, and cross-functional collaboration**, with experience delivering production features serving **1M+ users**.

EXPERIENCE

iOS App Developer, Target Corporation

Aug 2022 – Jul 2024

- As one of two engineers on the Registry sub-team, solely owned development and maintenance of **18 - 20 registry pages** on Target's iOS application serving **1M+ concurrent users**.
- Independently delivered the Registry Checklist feature end-to-end, driving **32K+ daily visits** and boosting customer engagement.
- Led **UIKit** → **SwiftUI migration** across **9–12 screens** with full unit test coverage, reducing crash rates and improving page load time by **18%** in a high-traffic production environment.
- Collaborated with product managers, designers and cross-functional engineering teams throughout Agile sprint cycles to deliver customer-facing features and contributed to code testing, reviews, beta releases and production support to ensure application quality and reliability.
- Delivered features across every release cycle while implementing asynchronous RESTful APIs, JSON parsing and MVVM architecture patterns.
- **Tech Stack:** Swift, UIKit, REST APIs, JSON, Git, Agile, XCode

Research Engineer, UNC Charlotte – MetaBo Food Vision Pro

Nov 2025 – May 2026

- Developing a spatial computing application for Apple Vision Pro that delivers personalized nutritional recommendations based on user health profiles and disease-specific dietary needs.
- Leveraging a dataset of **32K+ food products** and spatial imagery to identify grocery items and generate context-aware nutritional guidance.
- Building 3D object integration and interactive visionOS workflows for real-time grocery item analysis and product comparison.
- **Tech Stack:** SwiftUI, VisionOS, Reality Composer Pro, REST APIs

Instructional Assistant, UNC Charlotte

Aug 2025 – May 2026

- Supported **195 students** across three course sections through recitations, office hours, grading, and Tableau-based data visualization guidance.
- Assisted with exam preparation and provided individualized academic support to improve student understanding and assignment quality.
- **Tech Stack:** Tableau, Data Visualization, Excel

Risk Analyst Intern, PwC

Jan 2022 – Jul 2022

- Applied machine learning techniques to analyze **100K+ client records**, identifying fraud patterns and supporting risk mitigation initiatives.
- Automated analytical workflows and data validation processes to improve reporting efficiency and decision-making.
- **Tech Stack:** Python, Pandas, Machine Learning, Data Analysis

PROJECTS

Visual Speech-Aware 3D Facial Reconstruction

Jan 2025 – May 2025

- Built speech-aware 3D facial reconstruction using SPECTRE/DECA backbone with MobileNetV2 and temporal convolution, replacing ResNet for expression and jaw pose prediction.
- Trained on **LRS3 dataset (118K+ synchronized audio-video clips)**; evaluated using CER, WER and Viseme error rate; deployed the complete pipeline on AWS EC2 with Streamlit frontend.
- **Tech Stack:** Python, PyTorch, OpenCV, AWS, ngrok

Smart Voting System – Published Research ICICCT 2022, Springer

Jun 2021 – Jul 2022

- Developed a biometric fingerprint recognition using Gabor filtering and Hamming distance-based minutiae matching.
- **Tech Stack:** Python, OpenCV, Pandas, GSM Module

EDUCATION

University of North Carolina at Charlotte

May 2026

Master of Science in Computer Science

GPA: **3.9/4.0**

Amrita Vishwa Vidyapeetham, India

Aug 2022

Bachelor of Technology in Computer Science

TECHNICAL SKILLS

Languages: Swift, Python, Java, JavaScript, Kotlin, C, C++, HTML/CSS

Frameworks & Tools: Xcode, Android Studio, SwiftUI, UIKit, visionOS, Reality Composer Pro, PyTorch, OpenCV, Scikit-learn, React, Node.js, Docker, Git, CI/CD, Tableau, Snowflake

Databases: PostgreSQL, MySQL, SQLite

Cloud: AWS Data Engineering Certified